

[← Back](#)

10 marks

$$W \rightarrow 1$$

input

```
hello world
```

SIMATS Online Lab

Convert JPG to PDF, Images JP

harsha298250/PYTHON-CSA06

Ask Gemini

Verify that it's you

Not secure

simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=246

SIMATS

ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

M Harshavardhan

192510173RC

Length Encoding

10 marks

Q4

Advanced Text Statistics

Analyser

10 marks

4/4 answered

• Dictionary or manual counting

• Sorting using loops or sorted()

Input: Hello World

output:
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1

Your Code

```
1 str=input("enter the character\n")
2 count={}
3 for ch in str:
4     if ch!=" ":
5         if ch in count:
6             count[ch]+=1
7         else:
8             count[ch]=1
9 for ch in sorted(count):
10     print(ch,'->',count[ch])
```

Submitted

Input

hello world

Output

```
enter the character
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
```

© 2026 SIMATS Online Lab. All rights reserved.

SIMATS Online Lab

Convert JPG to PDF, Images JP

harsha298250/PYTHON-CSA06

Ask Gemini

Not secure

simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=246

Verily that it's you

SIMATS

ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

M Harshavardhan

19/25/11719C

Coding Challenge

← Back

QUESTIONS

Q1

String Character Frequency Counter

10 marks

Q2

Vowel and Consonant Classifier with Statistics

10 marks

Q3

String Compression and Run-Length Encoding

10 marks

Q4

Advanced Text Statistics Analyzer

10 marks

4/4 answered

Q2 Vowel and Consonant Classifier with Statistics

10 marks

Accept a sentence from the user and display: (a) total vowels and consonants, (b) the most frequent vowel, (c) the least frequent consonant, and (d) the ratio of vowels to consonants as a decimal rounded to 2 places.

Constraints:

- Case-insensitive analysis.
- Ignore spaces and digits.
- Handle sentences with no vowels or no consonants gracefully.

Skills Assessed:

- Character classification with conditionals.
- Counting and comparison logic
- Arithmetic and rounding

Sample Input :

Python Programming

Expected Output : Vowels: 5 | Consonants: 12 Most frequent vowel: o (2) Least frequent consonant: y (1) Vowel-to-Consonant Ratio: 0.42

Your Code

```
1 str=input("Enter the sentence:")
2 vowel="aeiou"
3 vc=0
4 cc=0
5 vf={}
6 cf={}
7 for ch in str:
8     if ch.isalpha():
9         if ch in vowel:
10             vc+=1
11             if ch in vf:
12                 vf[ch]+=1
```

Input

hibro

Output

Output

[← Back](#)

10 marks

- Single characters without count still output as single char.
- Handle empty strings.
- Decompressor should handle all valid RLE strings.

Your Code

Input

abc

Output

SIMATS Online Lab

Convert JPG to PDF, Images JP...

harsha298250/PYTHON-CSA06

Ask Gemini

Not secure

simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=246

Verily that it's you

SIMATS

ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

M Harshavardhan

1925101719C

Your Code

```
1 s=input("enter the characters:")
2 q=""
3 count=1
4 for i in range(len(s)):
5     if i<len(s)-1 and s[i]==s[i+1]:
6         count+=1
7     else:
8         q+=s[i]
9         if count>1:
10            c+=str(count)
11            count=1
12 if len(c)<len(s):
13     print("\ncompressed:",c)
14 else:
15     print("\ncompression not beneficial")
16     print("original:",s)
17 d=""
18 i=0
19 while i< len(c):
20     ch=c[i]
21     i+=1
22     if i<len(c) and c[i].isdigit():
23         d+=ch*int(c[i])
24         i+=1
25     else:
26         d+=ch
27 print("decompressed",d)
```

Submitted

Input

Output

```
enter the characters:
compression not beneficial
original: abc
decompressed abc
```

© 2026 SIMATS Online Lab. All rights reserved.

[← Back](#)

Q4 Advanced Text Statistics Analyser

10 marks

Q4 

Advanced Text Statistics
Analyser

10 merke

4/4 answered

Flesch Reading Ease: 58.3 (Standard)

Your Code

```
1 text=input("enter the text:")
2 words=text.split()
3 print ("words:",len(words))
4 sentence=text.count(".") + text.count("!") + text.count("?")
5 if sentence==0:
```

Input

```
python is a powerful programming
language
```

```

1 text=input("Enter the text:")
2 words=text.split()
3 print ("words:", len(words))
4 sentence=text.count(".") + text.count(",") + text.count("?")
5 if sentence==0:
6     sentence=1
7 print ("sentence:", sentence)
8 print ("paragraphs: 1")
9 letters=0
10 for w in words:
11     letters+=len(w)
12 avgwords=letters/len(words)
13 print ("average word length:", round(avgwords,2))
14 avgsentence=len(words)/sentence
15 print ("average sentence length:", round(avgsentence,2))
16 stop=["the", 'is', 'a', 'in', 'an', 'of', 'and', 'to', 'for']
17 freq={}
18 for w in words:
19     w=w.lower().strip(",.!?")
20     if w not in stop:
21         if w in freq:
22             freq[w]+=1
23         else:
24             freq[w]=1
25 print ("top 5 words:")
26 count=0
27 while count <5 and len(freq)>0:
28     m=max(freq, key=freq.get)
29     print(m, ":", freq[m], ":", sep="")
30     del freq[m]
31     count+=1
32 vowels="aeiou"
33 syllables=0
34 for w in words:
35     w=w.lower()
36     new=1

```

python is a powerful programming language

```
python(1)
powerful(1)
programming(1)
language(1)
flesch reading ease: 31.55
```

```
11 letters=len(w)
12 avgwords=letters/len(words)
13 print("average word length:",round(avgwords,2))
14 avgsentence=len(words)/sentence
15 print("average sentence length:",round(avgsentence,2))
16 stops=["the","is","a","in","an","of","and","to","for"]
17 freq={}
18 for w in words:
19     w=w.lower().strip(",.!")
20     if w not in stops:
21         if w in freq:
22             freq[w]+=1
23         else:
24             freq[w]=1
25 print("top 5 words:")
26 count=0
27 while count<5 and len(freq)>0:
28     m=max(freq,key=freq.get)
29     print(m,"(",freq[m],")",sep=" ")
30     del freq[m]
31     count+=1
32 vowels="aeiou"
33 syllables=0
34 for w in words:
35     w=w.lower()
36     prev=False
37     for ch in w:
38         if ch in vowels:
39             if prev==False:
40                 syllables+=1
41             prev=True
42         else:
43             prev=False
44 score=206.835-1.015*(len(words)/sentence)-84.6*(syllables/len(words))
45 print("flesch reading ease:",round(score,2))
46
```